

Michael Bailey

AI Engineer — LLM Applications, Python & TypeScript

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PROFESSIONAL SUMMARY

Software engineer with 3 years at Sky, now leading AI adoption in the Global Reliability team. Built solo an AI-powered self-serve dashboard that replaces a BA-mediated change-request workflow consuming ~95 engineer-days/year: users describe a chart in natural language and drop returned widgets into their own dashboards, backed by a custom Model Context Protocol (MCP) server and agent orchestrator over OpenAI GPT. Previously owned Sky's internal monitoring dashboards — Next.js, GraphQL, OpenSearch — used across multiple departments. Software Developer Level 4 apprenticeship, Distinction.

KEY SKILLS

AI & LLM Applications: LLM application development, OpenAI (GPT) models, agentic workflow design, Model Context Protocol (MCP), custom agent orchestration, tool use / function calling, prompt engineering, streaming responses (SSE), embeddings, vector search (OpenSearch)

Languages: Python, TypeScript (confident), JavaScript · Java, Golang, C# (familiar)

Frameworks & Platforms: Next.js, React, Node.js, FastAPI, Express, GraphQL, Recharts

Data & Search: OpenSearch (search, analytics, vector), MySQL, SQL

Cloud: Google Cloud Platform (GCP)

CI & Tooling: Jenkins (CI/CD), Docker, pytest, Jest; deployed on Sky's internal FluidCloud platform

Ways of Working: Technical leadership of AI initiatives · running internal AI onboarding sessions · production incident response (Global Reliability) · stakeholder demos across Sky teams and to Comcast executives

EXPERIENCE

Software Engineer — Sky UK

Sep 2022 – Present · London · Global Reliability

AI initiatives led:

- Designed and built solo an AI-powered self-serve monitoring dashboard: users ask questions like *"show me 4G usage split by tech in a stacked bar chart over the last eight months"* and a custom agent returns chart widgets they can resize, edit in-place, and assemble into shareable, per-user dashboards — collapsing a BA-mediated change-request cycle of 1-2 days per chart into sub-minute self-serve.
- Architected a custom MCP server (TypeScript) and agent orchestrator (Python, uvicorn) to work around tool-calling limits in Sky's internal AI gateway — wiring OpenSearch data sources, a Next.js / React / TypeScript frontend, and OpenAI GPT models into a single tool-use loop, Dockerised and deployed via Jenkins on Sky's FluidCloud platform.
- Authored the compliance sign-off business case — covering security posture (prompt-injection boundaries, handle-based result retrieval to prevent raw-row leakage, tenant isolation), efficiency modelling (~35% projected gain against the ~95-day baseline), cost projection (~£100/month run-cost), and phased rollout.
- Ran 3 stakeholder demos during pre-rollout hardening: Sky's data engineering team, another Sky observability team, and a group of Comcast executives (Sky's parent company).
- Leading a department-wide shift to agentic workflows at my director's request — running onboarding sessions on using AI effectively and safely across engineering and non-engineering work.

Core engineering:

- Shipped an earlier reliability-insights tool for Sky's on-call rota — embedding-based retrieval over OpenSearch surfaces similar past incidents during triage, shortening time-to-context on repeat-class failures.

- Owned a suite of full-stack internal monitoring dashboards used across multiple Sky departments for 3 years (Next.js / React / TypeScript frontend, GraphQL middleware, OpenSearch data) — the direct foundation for the AI-powered self-serve dashboard above, re-using the same data contract.

EDUCATION

Software Developer (Level 4) Apprenticeship — Multiverse / Sky UK

Sep 2022 – Apr 2024 · Distinction

Completed the Software Developer Level 4 apprenticeship standard (Institute for Apprenticeships and Technical Education) while working full-time in Sky's engineering organisation. Training provider: Multiverse. End-point assessment awarded Distinction on 24 April 2024. Work-based project: Sky's internal monitoring dashboards (Next.js / React, GraphQL, OpenSearch).